

Experiment 02

Aim: To Prepare SRS in IEEE format for any case study.

Project Title: Hostel Management System.

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document describes the Software Requirements Specification (SRS) for a **Hostel Management System**. The system will manage student registration, room allocation, hostel records, fee management, complaints, and administrative activities efficiently.

1.2 Scope

The **Hostel Management System** provides an online platform for managing hostel operations. It allows students to register, apply for hostel accommodation, check room details, pay hostel fees, and submit complaints. The administrator can manage rooms, students, fee records, complaints, and hostel information.

1.3 Definitions, Acronyms, Abbreviations

- **SRS:** Software Requirements Specification
- **Admin:** Hostel administrator who manages hostel records
- **DB:** Database
- **HMS:** Hostel Management System

1.4 References

- IEEE 830-1998 Software Requirements Specification Standard
- College Hostel Rules and Regulations
- Sample Hostel Management Systems

1.5 Overview

This **SRS** describes the overall description, functional requirements, non-functional requirements, constraints, assumptions, interfaces, and supporting information for the **Hostel Management System**.

2. Overall Description

2.1 Product Perspective

The **Hostel Management System** is a standalone web application that automates hostel management processes. It can later be integrated with the college ERP system for student data synchronization.

2.2 Product Functions

- Student registration and login
- Hostel room allocation
- Hostel fee management
- Complaint registration and tracking
- Room availability display
- Student profile management
- Admin management of students, rooms, and fees
- Notification of hostel updates

2.3 User Characteristics

Students:

- Basic computer and internet knowledge
- Can apply for hostel, pay fees, and submit complaints

Admin:

- Hostel office staff with basic computer knowledge
- Manages rooms, students, fee records, and complaints

2.4 Constraints

- Must follow standard web security practices
- Compatible with popular web browsers
- Only authorized administrators can access management features
- Must comply with institutional data privacy policies

2.5 Assumptions & Dependencies

- Reliable internet connection is available
- Server hosting is provided by the institution
- Student details are provided by the college administration
- Online payment gateway is available for fee payment

3. Specific Requirements**3.1 Detailed Functional Requirements (Working Criteria)**

- Students must be able to register and log in securely.
- Students must be able to apply for hostel accommodation.
- The system must allocate rooms based on availability.
- Students must be able to view room details.
- Students must be able to pay hostel fees online.
- The system must store payment details in the database.
- Students must be able to submit complaints.
- The admin must be able to log in securely.
- The admin must be able to add, update, or remove student records.
- The admin must be able to allocate or change hostel rooms.
- The admin must be able to manage hostel fee records.
- The admin must be able to view and resolve complaints.
- The system must generate reports for students, rooms, and fee payments.

3.2 Non-functional Requirements

Performance:

- Pages should load within 3 seconds.

Storage:

- The system must store student records, room details, complaints, and payment information securely.

Security:

- Secure login for admin and students.
- Passwords must be encrypted.
- Student information must be protected.

Usability:

- User-friendly interface for students and administrators.

Interoperability:

- Compatible with major web browsers such as Chrome, Firefox, Edge, and Safari.

Flexibility:

- New hostel buildings and room categories can be added in the future.

Disaster Recovery:

- Regular database backups must be maintained to prevent data loss.

3.3 Interfaces

User Interface:

- Student Login Page
- Student Dashboard
- Hostel Application Form
- Fee Payment Page
- Complaint Registration Page
- Admin Dashboard

Hardware Interface:

- Standard web server
- Desktop, Laptop, Tablet, or Mobile device

Software Interface:

- Apache Web Server
- MySQL Database
- PHP/Java/Python (Backend)
- HTML, CSS, JavaScript (Frontend)

Communication Interface:

- Internet connection
- Email notifications for hostel approval and fee confirmation
- SMS notifications (optional)

4. Appendices

- Sample Student Registration Form
- Sample Hostel Application Form
- Sample Fee Receipt
- Sample Complaint Form
- Hostel Room Allocation Chart
- Glossary